

Apportionment: Hamilton, Webster, Huntington-Hill, and Dean get in line!

In this activity you will treat apportionment methods as functions from a continuous *quota space* to an *apportionment space* (sometimes called an *allocation space*),

$$F : \mathcal{Q} = \{(a, b) \in \mathbb{R}_{\geq 0}^2 : a + b = M\} \text{ to } \mathcal{M} = \{(k, \ell) \in \mathbb{Z}_{\geq 0}^2 : k + \ell = M\}$$

to discrete integer allocations. You will visualize apportionment methods as geometric partitions of the line segment $\mathcal{Q}$ that determine how to apportion the seats. That is, you're going to find $F^{-1}(\mathbf{m})$ for each $\mathbf{m} \in \mathcal{M}$.

Warm-Up: Two Groups, Five Seats. Before working in triangles and three dimensions, we begin with the simplest possible case: two groups sharing a fixed number of seats.

Suppose two groups, A and B, are dividing 5 seats. If group A receives a seats, then group B receives $5 - a$. So every possible allocation corresponds to a point on the line segment

$$0 \leq a \leq 5.$$

In this activity, you will compare Hamilton's method and Webster's method in this one-dimensional setting.

Quota Space on a Line. Think of a as the (possibly fractional) quota for group A. The quota for group B is automatically $5 - a$.

Checkpoint 1 Mark the integer points on the number line from 0 to 5.

What do these integer points represent in terms of seat allocations?

Solution. The integer points are 0, 1, 2, 3, 4, 5. The point k represents the allocation "A gets k seats and B gets $5 - k$ seats," i.e. $(k, 5 - k)$.

Checkpoint 2 What does an integer like $a = 2.7$ (or the ordered pair $(2.7, 2.3)$) represent in this setting?

Solution. It represents a quota (ideal fractional allocation) where A "deserves" about 2.7 seats and B about $5 - 2.7 = 2.3$ seats. An apportionment method will turn this into an integer allocation near $(3, 2)$ or $(2, 3)$. Which one depends on the method.

Hamilton's Method on the Line. Hamilton's method works by first taking lower quotas and then assigning leftover seats based on the size of the fractional parts.

In the two-group case, this process simplifies dramatically!

Checkpoint 3 Suppose group A has quota a .

- Write the lower quota for group A.
- Write the lower quota for group B.
- How many seats remain to be assigned after taking floors?

Solution. (a) A's lower quota is $\lfloor a \rfloor$. (b) B's lower quota is $\lfloor 5 - a \rfloor$.

(c) The remaining seats are $5 - (\lfloor a \rfloor + \lfloor 5 - a \rfloor)$. For non-integer a , we have $\lfloor a \rfloor + \lfloor 5 - a \rfloor = 4$, so there is 1 leftover seat. (If a is an integer, there are no leftover seats.)

Checkpoint 4 Show that Hamilton's method gives the extra seat to group A exactly when

$$a - \lfloor a \rfloor \geq (5 - a) - \lfloor 5 - a \rfloor.$$

Simplify this inequality.

Checkpoint 5 Find the value of a where the seat allocation switches.

Draw this point on the number line and label the intervals where Hamilton assigns:

- 2 seats to A,
- 3 seats to A.

Webster's Method on the Line. Webster's method uses a scaling factor (the divisor) and then rounds to the nearest integer.

In the two-group case, Webster's behavior becomes especially simple.

Checkpoint 6 Suppose Webster uses the quota a directly. Write the condition under which group A is rounded to 3 seats.

Express your answer as an inequality involving a .

Checkpoint 7 Find the value of a where Webster switches from giving A 2 seats to giving A 3 seats.

Mark this point on the same number line you used for Hamilton.

Huntington–Hill (Equal Proportions) on the Line. Now we add one more method, often called *Huntington–Hill* or *Equal Proportions*. In this warm-up we will describe it as a next-seat rule.

We will use the common convention that each group must receive at least 1 seat. So with 5 total seats and two groups, the possible final allocations are $(1, 4)$, $(2, 3)$, $(3, 2)$, $(4, 1)$.

Step 1: A next-seat score. Suppose a group has population p and currently has n seats. Its score for receiving the next seat is

$$\text{score}(p, n) = \frac{p}{\sqrt{n(n+1)}}.$$

The next seat goes to the group with the larger score. (If there is a tie, use a tie-breaking rule.)

Checkpoint 8 Starting from $(1, 1)$, there are 3 more seats to give out. Make a small table of the scores that would be compared at each step:

- When comparing A goes from 1 to 2 versus B goes from 1 to 2.
- When comparing A goes from 2 to 3 versus B goes from 1 to 2.
- When comparing A goes from 3 to 4 versus B goes from 1 to 2.

Solution. The score for going from n to $n + 1$ is $p/\sqrt{n(n+1)}$.

Comparing A: $1 \rightarrow 2$ versus B: $1 \rightarrow 2$ compares $\frac{p_A}{\sqrt{1 \cdot 2}}$ to $\frac{p_B}{\sqrt{1 \cdot 2}}$, so the first extra seat goes to whichever population is larger.

Later comparisons involve different denominators: A: $2 \rightarrow 3$ uses $\sqrt{2 \cdot 3} = \sqrt{6}$, while B: $1 \rightarrow 2$ uses $\sqrt{2}$, etc.

Step 2: Where does the outcome switch? As before, let a be A's quota on the line, so B's quota is $5 - a$. The Huntington–Hill method partitions the line into intervals where A receives 1, 2, 3, or 4 seats.

Checkpoint 9 The middle switch between A getting 2 seats versus 3 seats happens at $a = 2.5$. (This is the symmetric point where A and B have equal population.)

The other two switch points occur when the population ratio reaches $\sqrt{6}$ (or its reciprocal). Use the score comparison to show that the boundary between outcomes (1, 4) and (2, 3) occurs when

$$\frac{p_A}{\sqrt{2}} = \frac{p_B}{\sqrt{6}}.$$

Then solve for the corresponding quota value a .

Checkpoint 10 Use the switch points to label the intervals on the line where A receives 1, 2, 3, or 4 seats under Huntington–Hill.

Dean's Method on the Line. Dean's method is another divisor method, similar in structure to Webster's and Huntington–Hill's. The difference lies in how rounding is decided.

Instead of rounding to the nearest integer or using geometric means, Dean's method uses the *harmonic mean* as the cutoff between adjacent integers.

Step 1: Dean's Rounding Rule. Suppose a quota value q lies between two integers n and $n + 1$. Dean's method rounds:

- down to n if q is less than the harmonic mean of n and $n + 1$,
- up to $n + 1$ if q is greater than that harmonic mean.

Recall that the harmonic mean of n and $n + 1$ is

$$H(n, n + 1) = \frac{2n(n + 1)}{2n + 1}.$$

Checkpoint 11 Compute the harmonic mean cutoff between:

- 1 and 2
- 2 and 3

Step 2: Dean's Switching Points on the Line. As before, let a represent A's quota on the line with total 5 seats. Dean's rounding thresholds determine when A switches from receiving 2 seats to 3 seats, and from 1 to 2, etc.

Checkpoint 12 Using the harmonic mean cutoff between 2 and 3, determine the value of a where Dean switches from assigning A 2 seats to assigning A 3 seats.

Checkpoint 13 Use the harmonic mean cutoffs to find the approximate switch points where A moves:

- from 1 seat to 2 seats,
- from 3 seats to 4 seats.

Solution. Between 1 and 2 seats, the cutoff is $4/3 \approx 1.333$. So A switches from 1 to 2 seats at $a \approx 1.333$.

By symmetry, the cutoff between 3 and 4 seats occurs at $5 - 4/3 = 3.6667$.

Checkpoint 14 Label the intervals on the number line where Dean assigns A:

- 1 seat
- 2 seats
- 3 seats
- 4 seats

Comparing All Four Methods.

Checkpoint 15 Compare the switching points for Hamilton, Webster, Huntington–Hill, and Dean for the transition between 2 seats and 3 seats.

Arrange the methods from earliest switch to latest switch as a increases.

Checkpoint 16 In words: what does Dean’s method favor compared to Webster and Huntington–Hill?